

Idea Discovery & Validation — Deep Dive

Food Waste / Smart Fridge Tracker

1. The Problem (stated precisely)

People throw away food regularly because nothing tells them what's about to expire, and they don't know what they can cook with what they already have at home. This is a genuinely universal problem — it happens in every country, every household, regardless of income level. It's also an unusually **guilt-heavy** problem: people feel bad wasting food and money, which makes them more motivated to actually use a solution.

2. Who Has This Problem

Primary persona: “Sara, a university student living alone or with roommates”

- Buys groceries, forgets what's in the fridge, finds spoiled vegetables a week later
- Doesn't plan meals around what's already there, so buys duplicates

Secondary persona: “A household/family”

- Parents managing a shared fridge for the whole family, kids/spouse not tracking what's used
- Shared-household confusion: “who finished the milk?” “is this still good?”

Why reachable: university students, young professionals, and households are everywhere — but a smart way to reach them early would be through student communities (like my own university) or local social media groups.

3. Why This Is Still Unsolved

- Most existing apps require **manual data entry** (typing in every item, its expiry date) — which is tedious enough that people stop using the app after a few days
- Barcode-scanning apps exist but are clunky and don't work for fresh produce (a tomato doesn't have a barcode)
- The 2026 shift that makes this newly buildable: AI image recognition can now identify groceries from a simple photo, removing the tedious manual entry step entirely

4. Existing Solutions & Why They Fall Short

Existing tool	Why it doesn't work well
Manual grocery-tracking apps	Too tedious, people stop using them within days
Barcode scanner apps	Doesn't work for fresh produce, still manual for most items
Generic recipe apps	Not connected to what's actually in your fridge
Sticky notes / memory	What almost everyone actually does — unreliable

5. Proposed Solution

An app where you take a photo of your groceries (or fridge) once, and the AI:

1. Identifies what items you have
2. Estimates realistic expiry timelines (a banana is not the same as a jar of pickles)
3. Sends a gentle reminder before things go bad (“Your spinach expires in 2 days”)
4. Suggests a simple recipe using what’s about to expire, so it actually gets used

What makes this different: it removes the manual typing that kills every existing app — the photo does the work.

6. Core MVP Features (3-5, kept realistic)

1. **Photo-based item entry:** snap a photo of groceries, AI identifies items and adds them to your list
2. **Smart expiry estimates:** reasonable default expiry windows per food type (editable if wrong)
3. **Expiry reminders:** simple notification a day or two before something goes bad
4. **“What can I cook?” suggestions:** basic recipe matching using what’s expiring soon
5. *(Optional v2)* Shared household mode — multiple people update the same fridge list

7. User Journey (step-by-step)

1. User comes home from grocery shopping, opens the app, takes one photo of their bags/fridge
2. AI identifies items (“tomatoes, milk, spinach, eggs...”) and adds them with estimated expiry dates
3. Two days before spinach expires, app sends a reminder plus a simple recipe idea using spinach
4. User marks items as “used” or “thrown out” so the list stays accurate

8. Tech Stack (realistic for a student to build)

- **Frontend:** React + TypeScript + Vite (same stack as my portfolio website)
- **Image recognition:** an AI vision model (like Claude or a food-

recognition API) to identify groceries from photos

- **Backend/storage:** Supabase or Firebase for storing each user's fridge list
- **Notifications:** simple browser or email reminders for MVP (no need for a full mobile app push system yet)

9. Data & Privacy Considerations

- Only the user's own food photos and list — nothing sensitive
- No need to store payment or personal identity info for the MVP

10. Risks & Honest Challenges

- **AI misidentifying food** — needs a simple “fix this” edit option so wrong guesses don't frustrate users
- **Habit-forming is hard** — if people forget to photograph their groceries, the app becomes useless; reminders to “scan your groceries” after a shopping trip help
- **Expiry estimates vary** — a “best guess” system with easy manual editing is safer than pretending to be perfectly accurate

11. How to Validate Before Building

1. Ask 10 people (classmates, family) how they currently track what's in their fridge — most will say “I don't” or “I forget things and throw them out”
2. Manually test the idea: for one week, take photos of your own groceries and try to guess expiry plus suggest what to cook — see if it actually changes your habits
3. If people say “I'd actually use this” unprompted, that's validation

12. Success Metrics for v1

- Reduction in self-reported food thrown away per week
- How often users actually open the app after grocery shopping (habit signal)
- How often the “what can I cook” suggestion actually gets used